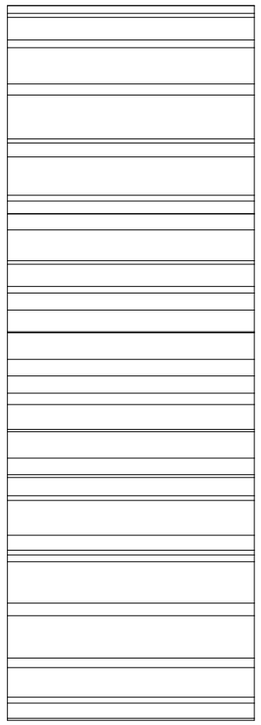
# VAYN shoulder · milling drawing · 36 Planet-2

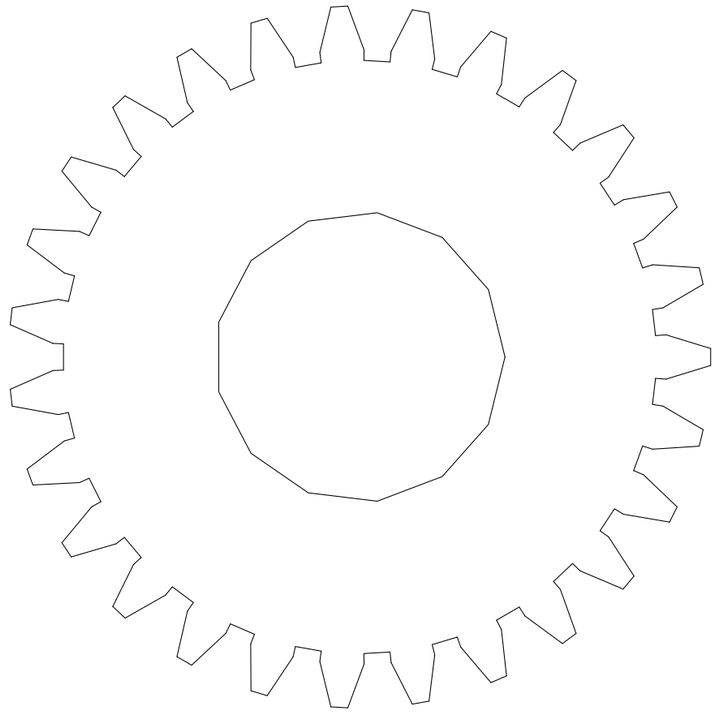


FRONT · Z horizontal, +X up



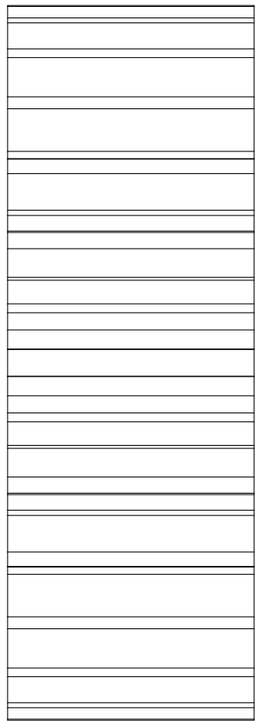
scale 4.03:1 view span 10.0 × 28.9 mm

END · from the arm side (+Z)



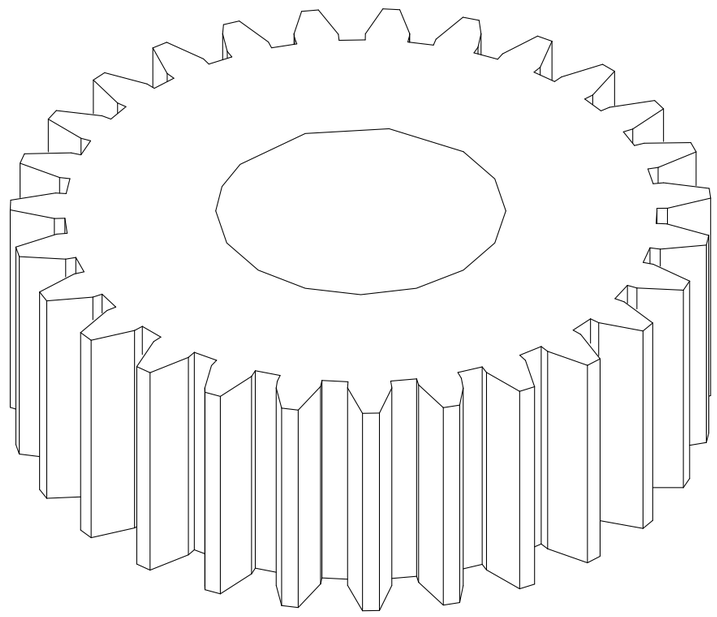
scale 3.95:1 view span 28.9 × 29.0 mm

SIDE · Z horizontal, +Y up



scale 4.02:1 view span 10.0 × 29.0 mm

ISOMETRIC



scale 3.94:1 view span 29.0 × 24.9 mm

## 36

### Planet-2

Quantity in this actuator: 3
Material: 20MnCr5
Group: Gears
Finished size: Ø29 × 10 mm face, bore Ø12 mm
Mass: 35.24 g each
Model bounding box, X × Y × Z: 28.9 × 29.0 × 10.0 mm
Raw stock: Same Ø35 bar as planet-1
Process: Turn, hob, part off
Heat treatment / finish: Carburize, case 0.15–0.20 mm, 58–62 HRC. Grind the bore after hardening.
STEP solid: 36.1 Planet-2 (20MnCr5)

**Controlled features**

- 27 teeth, module 1.0. Span over 4 teeth W4 = 10.687 mm, minus 0.015 / plus 0.
- ISO 1328 grade 7–8 is accepted. Do not grind the teeth. This is the high-load mesh, about 1,350 MPa at 30 N·m.
- Bore Ø12 N6.
- Three identical parts. In the STEP they are 36.1, 36.2 and 36.3.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 36_planet_2.pdf